H.T.No: | | | | | | | | | | |

Course Code: **231CS5T03**

B.Tech – V Semester End Examinations Supplementary (AR23) – Jan 2026

FORMAL LANGUAGES AND AUTOMATA THEORY
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 70**

1. Answer all 10 questions from Section-A. Each question carries 2 marks.
2. Answer one question from each unit in Section-B. Each question carries 10 marks.

SECTION –A **10 X 2 =20**

1	a	What is the primary need for studying Automata Theory in computer science?	L1	CO1	[2M]
	b	Define a Deterministic Finite Automaton (DFA) and list its components.	L1	CO1	[2M]
	c	What is a regular expression (RE)?	L1	CO1	[2M]
	d	What is the pumping lemma for regular sets?	L2	CO2	[2M]
	e	Define a Context-Free Grammar and list its components	L1	CO3	[2M]
	f	Differentiate between leftmost and rightmost derivations in CFGs.	L2	CO3	[2M]
	g	Define a Pushdown Automaton and list its key components.	L1	CO2	[2M]
	h	Describe the concept of Instantaneous Description in the context of PDA.	L2	CO2	[2M]
	i	Define a Turing Machine and list its key components.	L1	CO5	[2M]
	j	Differentiate between Universal and Restricted Turing Machines.	L2	CO5	[2M]

SECTION –B **5 X 10 =50**

UNIT-I

2	a	Explain the central concepts of Automata Theory and discuss how they contribute to the design of computational models.	L2	CO1	[5M]
	b	Design a DFA for a language that accepts binary strings ending in "01". Justify your design choices.	L3	CO2	[5M]

OR

3	a	Compare and contrast Mealy and Moore machines with examples. How do they differ in terms of output generation	L4	CO1	[5M]
	b	Discuss the process of converting an NFA into a DFA. Why is this conversion significant in automata theory?	L5	CO1	[5M]

UNIT – II

4	a	Convert the given regular expression (a+b)*ba into NFA	L2	CO3	[5M]
	b	Analyze the Pumping Lemma and its application in proving non-regularity. Use the Pumping Lemma to show that L = $\{a^n b^n\}$	L4	CO1	[5M]

OR

5	a	Evaluate the process of simplifying Regular Expressions using identity rules.	L5	CO1	[5M]
	b	Classify grammars based on the Chomsky Hierarchy and explain their significance.	L2	CO3	[5M]

(P.T.O)

UNIT – III

6	a	Define leftmost, rightmost derivations and parse tree. Give leftmost, rightmost derivations and parse tree for the string 'baabba' to the grammar $S \rightarrow bA \mid aB, A \rightarrow a \mid aS \mid bAA, B \rightarrow b \mid bS \mid aBB$.	L2	CO3	[5M]
	b	Explain the significance of formal languages in computational theory and programming language design. Then, determine whether the language $L = \{a^n b^n \mid n \geq 0\}$ is regular or context-free.	L2	CO3	[5M]
		OR			
7	a	Analyze the concept of ambiguous grammars and discuss their impact on parsing. Demonstrate ambiguity by proving that the grammar $E \rightarrow E + E \mid E * E \mid id$ generates multiple parse trees for the same expression.	L4	CO3	5M]
	b	Discuss closure properties of context-free languages and their implications.	L2	CO3	[5M]

UNIT – IV

8	a	Describe the formal definition and working model of a Pushdown Automaton (PDA). Then, design a PDA that accepts the language $L = \{a^n b^n \mid n \geq 1\}$, where the number of a's is equal to the number of b's.	L3	CO2	[5M]
	b	Compare Deterministic and Non-Deterministic PDAs with examples.	L4	CO4	[5M]
		OR			
9	a	Explain how a Pushdown Automaton (PDA) processes input using Instantaneous Descriptions. Then, trace the ID sequence for the input string aabb in a PDA that accepts the language $L = \{a^n b^n \mid n \geq 1\}$.	L3	CO4	[5M]
	b	Analyze the conversion process from CFG to PDA.	L4	CO4	[5M]

UNIT – V

10	a	Define a Turing Machine and explain its formal computational model.Construct a Turing Machine that accepts the language $L = \{a^n b^n c^n \mid n \geq 1\}$, where the number of a's, b's, and c's are equal and appear in sequence.	L3	CO5	[5M]
	b	Describe how Transition Tables and Diagrams represent TM behavior.	L3	CO5	[5M]
		OR			
11	a	Design a Turing Machine for a simple arithmetic operation.	L3	CO5	[5M]
	b	Explain about Classes of P and NP, NP- Hard and NP Complete problems.	L2	CO5	[5M]

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